

## BI / read / 12

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| query       | BI / read / 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |  |  |
| title       | How many persons have a given number of messages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |  |  |
| pattern     | <div><div><div>2. personCount = count</div><div><div>Person</div></div><div>count Persons grouped by messageCount value</div></div><div><div>«opt»<br/>hasCreator</div><div><div>1. messageCount = count</div><div><div>Message</div><div>content not empty and length &lt; \$lengthThreshold and \$startDate &lt; creationDate</div></div></div><div><div>replyOf*0..</div><div><div>Post</div><div>language in \$languages</div></div></div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |  |  |
| description | <p>For each Person, count the number of Messages they made (messageCount). Only count Messages with the following attributes:</p> <ul style="list-style-type: none"><li>• Its content is not empty (and consequently, the imageFile attribute is empty for Posts).</li><li>• Its creationDate is after \$startDate (exclusive, equality is not allowed).</li><li>• Its length is below the \$lengthThreshold (exclusive, equality is not allowed).</li><li>• It is written in any of the given \$languages.</li></ul> <ul style="list-style-type: none"><li>– The language of a Post is defined by its language attribute.</li><li>– The language of a Comment is that of the Post that initiates the thread where the Comment replies to.</li></ul> <p>The Post and Comments in the reply tree’s path (from the Message to the Post) do not have to satisfy the constraints for content, length, and creationDate.</p> <p>For each messageCount value, count the number of Persons with exactly messageCount Messages (with the required attributes).</p> |  |  |  |  |
| params      | <div><div>1</div><div>\$startDate</div><div>Date</div><div>Selected randomly from a 60-day interval.</div></div> <div><div>2</div><div>\$lengthThreshold</div><div>32-bit Integer</div><div>Balanced against startDate to filter around 30% of the Messages within a language and keep the variance low.<br/>The selection of this parameter uses a factor table of bucketed Message lengths and creation dates.</div></div> <div><div>3</div><div>\$languages</div><div>{String}</div><div>Only the most frequently used languages</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  |  |  |
| result      | <div><div>1</div><div>messageCount</div><div>32-bit Integer</div><div>A</div><div>Number of Messages created</div></div> <div><div>2</div><div>personCount</div><div>32-bit Integer</div><div>A</div><div>Number of Persons with messageCount Messages</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |  |  |
| sort        | <div><div>1</div><div>personCount</div><div>↓</div><div></div></div> <div><div>2</div><div>messageCount</div><div>↓</div><div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |  |  |
| limit       | n/a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  |  |  |
| CPs         | 1.1, 1.2, 1.4, 2.6, 3.2, 4.2, 4.3, 8.1, 8.2, 8.3, 8.4, 8.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |  |  |